Comparison of Monte Carlo Techniques for Estimating a 2D Normalizing Constant

December 9, 2025

1 Introduction

In this report, we consider the problem of numerically estimating the normalizing constant

$$Z = \int_{\mathbb{R}^2} \tilde{\rho}(x)\,\mathrm{d}x,$$

where the unnormalized density $\tilde{\rho} : \mathbb{R}^2 \to (0, +\infty)$ is given by

$$\tilde{\rho}(x) = \exp\left(-\frac{x_1^2}{2} - \frac{1}{2}\left(2x_2 + x_1^2 - 1\right)^2\right).$$

The quantity Z plays the role of a normalizing constant for the probability density $\rho(x) = \frac{1}{Z}\tilde{\rho(x)}$ and is not available in closed form. Consequently, numerical approximation is required.

In this report, we implement and compare two Monte Carlo techniques for estimating Z: Monte Carlo Integration with uniform sampling and Importance Sampling. We analyze the performance of each method in terms of accuracy and computational efficiency.

Our aim is to study how the estimation error behaves as the sample size N increases, evaluate the influence of the proposal distribution in importance sampling, and compare the efficiency of the two approaches in terms of variance and computational cost.

2 Truncation Error: Justification of Restricting to $[-4, 4]^2$

To make the numerical integration feasible, we restrict the domain of integration to the square $[-4, 4]^2$. We justify this truncation by estimating the contribution of the tails outside this region. Consider the tail integral

$$E = \int_{\mathbb{R}^2 \setminus [-4,4]^2} \tilde{\rho}(x)\,\mathrm{d}x.$$

Let A_1 denote the set $(\mathbb{R} \setminus [-4, 4]) \times \mathbb{R}$, and let A_2 denote the set $\mathbb{R} \times (\mathbb{R} \setminus [-4, 4])$. Since $\mathbb{R}^2 \setminus [-4, 4]^2 \subset A_1 \cup A_2$,

we have

$$E = \int_{\mathbb{R}^2 \setminus [-4,4]^2} \tilde{\rho}(x)\,\mathrm{d}x \le \int_{A_1} \tilde{\rho}(x)\,\mathrm{d}x + \int_{A_2} \tilde{\rho}(x)\,\mathrm{d}x.$$

We estimate each term on the right-hand side separately.

2.1 Estimating $\int_{A_1} \tilde{\rho}(x)\,\mathrm{d}x$

First, we separate $\tilde{\rho}$ into two parts as follows:

$$\tilde{\rho}(x) = \exp\left(-\frac{x_1^2}{2}\right) \cdot \exp\left(-\frac{1}{2}(2x_2 + x_1^2 - 1)^2\right).$$

By Tonelli's Theorem, we have

$$\int_{A_1} \tilde{\rho}(x)\,\mathrm{d}x = \int_{\mathbb{R}\setminus[-4,4]} \left(\exp\left(-\frac{x_1^2}{2}\right) \int_{\mathbb{R}} \exp\left(-\frac{1}{2}(2x_2 + x_1^2 - 1)^2\right)\,\mathrm{d}x_2\right)\,\mathrm{d}x_1.$$

The inner integral can be computed via the substitution $u = 2x_2 + x_1^2 - 1$, yielding

$$\int_{\mathbb{R}} \exp\left(-\frac{1}{2}(2x_2 + x_1^2 - 1)^2\right)\,\mathrm{d}x_2 = \frac{1}{2}\int_{\mathbb{R}} \exp\left(-\frac{u^2}{2}\right)\,\mathrm{d}u = \frac{\sqrt{2\pi}}{2}.$$

Thus, we have

$$\int_{A_1} \tilde{\rho}(x)\,\mathrm{d}x = \frac{\sqrt{2\pi}}{2} \int_{\mathbb{R}\setminus[-4,4]} \exp\left(-\frac{x_1^2}{2}\right)\,\mathrm{d}x_1.$$

The remaining integral can be bounded using standard Gaussian tail bounds:

$$\int_{\mathbb{R}\setminus[-4,4]} \exp\left(-\frac{x_1^2}{2}\right)\,\mathrm{d}x_1 = 2\int_4^\infty \exp\left(-\frac{x_1^2}{2}\right)\,\mathrm{d}x_1 \le 2 \cdot \frac{1}{4}\exp\left(-\frac{16}{2}\right) = \frac{1}{2}e^{-8}.$$

Therefore, we obtain the bound

$$\int_{A_1} \tilde{\rho}(x)\,\mathrm{d}x \le \frac{\sqrt{2\pi}}{4}e^{-8} = O(10^{-5}).$$

2.2 Estimating $\int_{A_2} \tilde{\rho}(x)\,\mathrm{d}x$

A fully rigorous bound requires analyzing the decay of the term

$$\exp\left(-\frac{1}{2}(2x_2 + x_1^2 - 1)^2\right)$$

as $|x_2| \to \infty$. Since

$$(2x_2 + x_1^2 - 1)^2 \ge (2|x_2| - |x_1^2 - 1|)^2,$$

we expect the integrand to decay at least Gaussianly in x_2. However, due to time limitations, we do not present a fully explicit tail bound here. Instead, we note that numerical evaluation of the integral on a larger box (We will discuss about it in Section 3.4) shows that the additional mass in the region $|x_2| > 4$ is several orders of magnitude smaller than the Monte Carlo sampling error for all sample sizes used in this report.

2.3 Conclusion

Combining the above observations, we conclude that

$$E = \int_{\mathbb{R}^2 \setminus [-4,4]^2} \tilde{\rho}(x)\,\mathrm{d}x$$

is negligible compared to the statistical error of the Monte Carlo estimators studied later. Thus restricting the integral to $[-4,4]^2$ is justified for practical numerical purposes.

3 Monte Carlo Integration with Uniform Sampling

In this section, we implement the Monte Carlo Integration method using uniform sampling over the square $[-4,4]^2$ to estimate the normalizing constant Z.

3.1 Method and Implementation

3.1.1 Monte Carlo Method

The Monte Carlo estimator for Z using uniform sampling is given by

$$\hat{Z}_N = \frac{\mu(\Omega)}{N} \sum_{i=1}^{N} \tilde{\rho}(X_i), \quad X_i \sim \mathcal{U}(\Omega),$$

where $\Omega = [-4,4]^2$ is the region of integration, X_i are i.i.d. samples drawn uniformly from the region Ω, $\mu(\Omega)$ is the Lebesgue measure of Ω (which equals 64 in this case), and N is the total number of samples.

By Strong Law of Large Numbers, $\hat{Z}_N$ converges to Z almost surely as $N \to \infty$. Additionally, Monte Carlo Method is unbiased, i.e., $\mathbb{E}[\hat{Z}_N] = Z$ for all N.

3.1.2 Theoretical Error Analysis

The error of the Monte Carlo method is defined as

$$e_N = \hat{Z}_N - Z.$$

The root mean square error (RMSE) is given by

$$\mathrm{RMSE} = \sqrt{\mathbb{E}[e_N^2]}.$$

Calculating directly, we have

$$\mathbb{E}[e_N^2] = \mathbb{E}\left[\frac{\mu(\Omega)^2}{N^2} \sum_{i,j} \left(\tilde{\rho}(X_i) - Z\right)\left(\tilde{\rho}(X_j) - Z\right)\right] = \frac{\mu(\Omega)^2}{N} \mathrm{Var}(\tilde{\rho}(X)),$$

where $\mathrm{Var}(\tilde{\rho}(X)) = \dfrac{1}{\mu(\Omega)} \displaystyle\int_\Omega (\tilde{\rho}(x) - \frac{Z}{\mu(\Omega)})^2 \, \mathrm{d}x$.
Thus, the RMSE can be expressed as

$$\mathrm{RMSE} = \frac{\mu(\Omega)}{\sqrt{N}} \sqrt{\mathrm{Var}(\tilde{\rho}(X))}.$$

This shows that the RMSE decreases at a rate of $O(N^{-1/2})$ as the sample size N increases.

3.1.3 Algorithm

The algorithm for Monte Carlo Integration with uniform sampling is as follows:

1. Define the region of integration $\Omega = [-4, 4]^2$ and compute its measure $\mu(\Omega) = 64$.
2. For a given sample size N, generate N i.i.d. samples $X_1, X_2, \ldots, X_N$ from the uniform distribution $\mathcal{U}(\Omega)$.
3. Compute the Monte Carlo estimator

$$\hat{Z}_N = \frac{\mu(\Omega)}{N} \sum_{i=1}^{N} \tilde{\rho}(X_i).$$

4. Return the estimate $\hat{Z}_N$.
5. Compute the empirical RMSE by repeating the estimation process multiple times for each N. (This requires knowing the true value of Z, which can be approximated using a large sample size.)
6. Repeat the above steps for different values of N to analyze the convergence behavior.

3.2 Numerical Experiments

We conduct numerical experiments to evaluate the performance of the Monte Carlo Integration method with uniform sampling.

3.2.1 Setup

We vary the sample size N from 10^2 to 10^7 and compute the estimator $\hat{Z}_N$ for each N. For each sample size, we repeat the estimation process 1000 times to compute the empirical RMSE. We generate i.i.d. samples $X_i \sim \mathcal{U}(\Omega)$ using NumPy's uniform random number generator.

Since the true value of Z is unknown, and the uniform Monte Carlo estimator exhibits large variance for moderate sample sizes, we compute a high-accuracy reference estimate by using a very large sample size ($N = 10^8$). The reference estimate Z_{ref} is used to approximate the true value of Z for RMSE calculations:

$$\mathrm{RMSE} \approx \sqrt{\frac{1}{K} \sum_{k=1}^{K} (\hat{Z}_N^{(k)} - Z_{\mathrm{ref}})^2},$$

where $K = 100$ is the number of repetitions for each N, and $\hat{Z}_N^{(k)}$ is the k-th estimate for sample size N. This allows stable error measurement despite the high variance of the uniform estimator.

3.2.2 Experiment Results

Figure 3.1 shows the convergence of the uniform Monte Carlo estimator $\hat{Z}_N$ as the sample size N increases. The curve exhibits the following key features:

- A large variance when N is small.
- Stabilization around $Z_{\text{ref}} = 3.13$.
- Visually convergence as N grows.

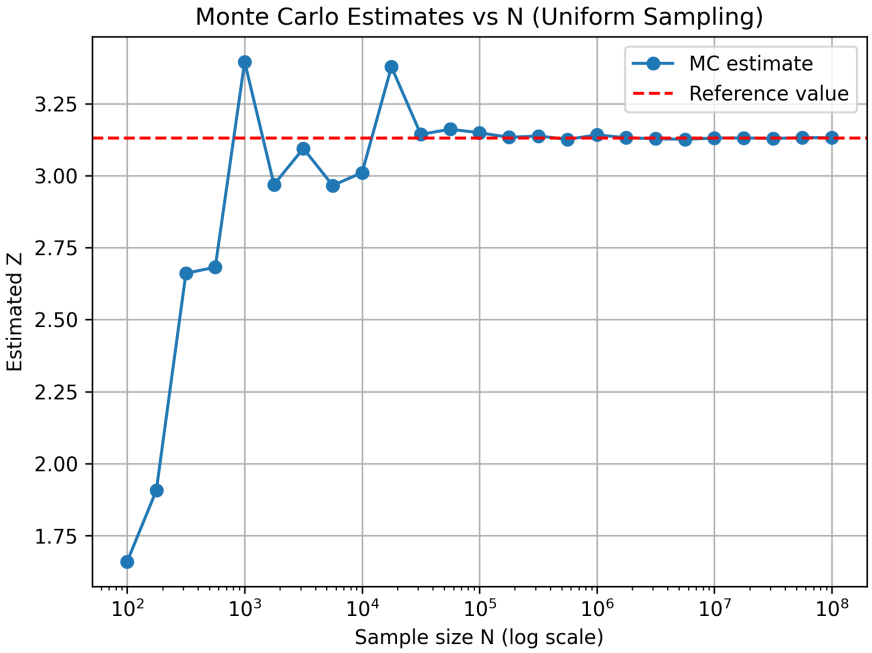


Figure 3.1: Convergence of Uniform Monte Carlo Estimator

Figure 3.2 displays the distribution of 3000 independent uniform Monte Carlo estimators with sample size $N = 5000$. The histogram exhibits:

- A roughly Gaussian shape, consistent with the Central Limit Theorem.
- A mean close to $Z_{\text{ref}} = 3.1331$.
- A wide spread, indicating high variance for this sample size.

Figure 3.3 presents the RMSE of the uniform Monte Carlo estimator as a function of sample size N on a log-log scale.

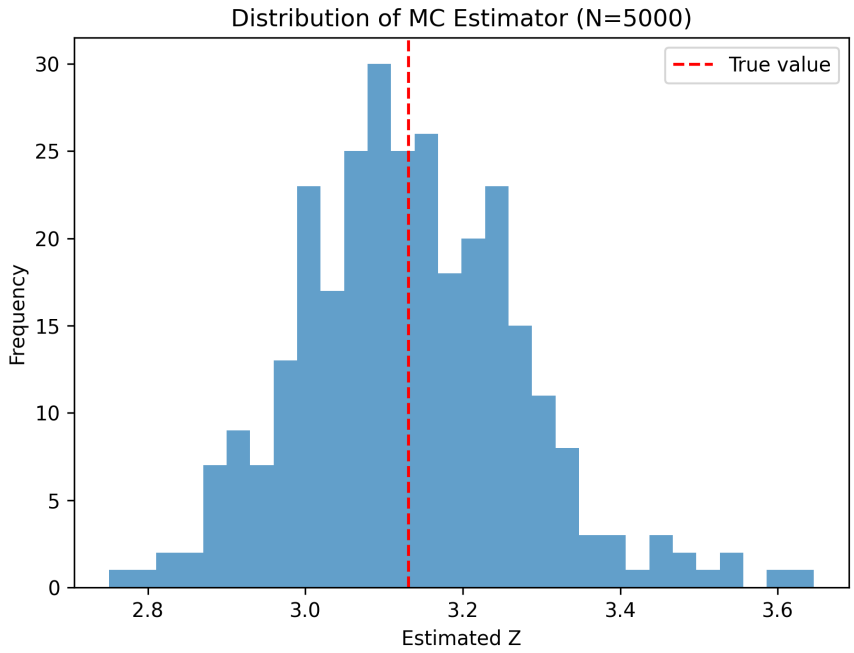


Figure 3.2: Distribution of Uniform Monte Carlo Estimator ($N = 5000$)

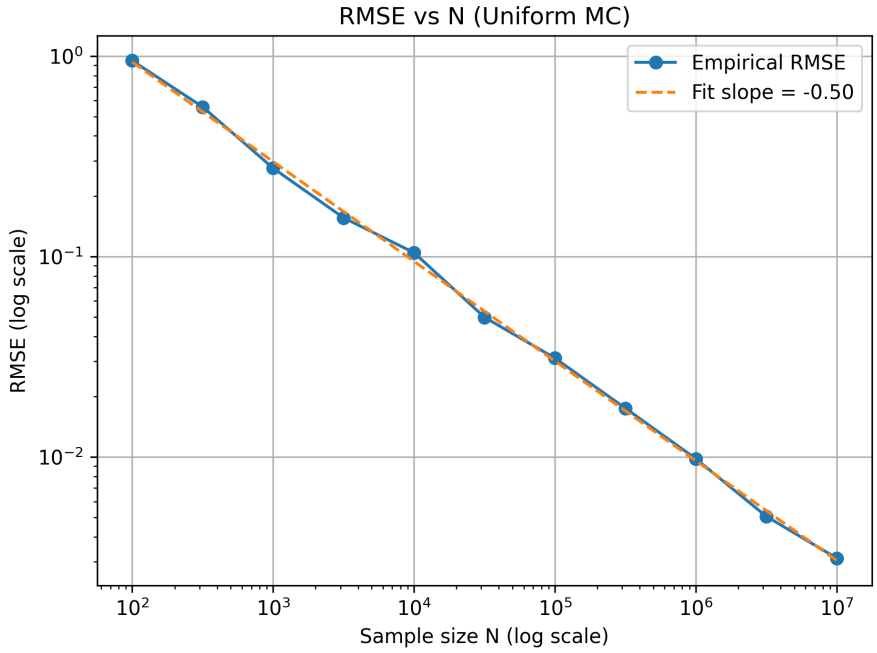


Figure 3.3: RMSE of Uniform Monte Carlo Estimator vs Sample Size

The RMSE plot demonstrates the expected $O(N^{-1/2})$ decay rate, confirming the theoretical analysis.

Figure 3.4 shows a scatter plot of 5000 uniform samples drawn from $\Omega = [-4, 4]^2$.

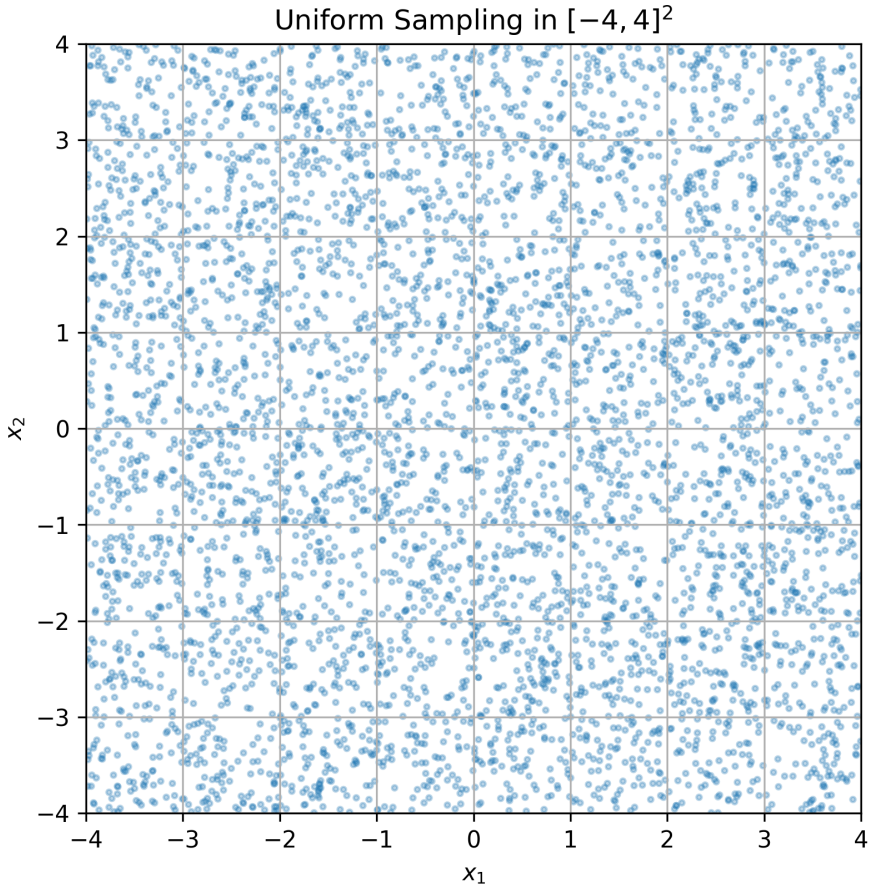


Figure 3.4: Sample Distribution of Uniform Monte Carlo Sampling ($N = 5000$)

3.3 Discussion

The numerical experiments validate the theoretical properties of the Monte Carlo Integration method with uniform sampling.

The estimator converges to the true value of Z as the sample size increases, and the RMSE decreases at the predicted rate of $O(N^{-1/2})$.

However, the high variance of the uniform estimator(As Figure 3.2 shows) for moderate sample sizes highlights the limitations of this approach: The large spread of estimates indicates that a very large number of samples is required to achieve high accuracy and the estimator converges slowly. This motivates the exploration of alternative methods, such as Importance Sampling,

3.4 Numerical Verification of Domain Restriction

To verify the justification of restricting the integration domain to $[-4, 4]^2$, we perform numerical experiments by estimating the integral over a larger box $[-8, 8]^2$. Using uniform Monte Carlo sampling with sample size

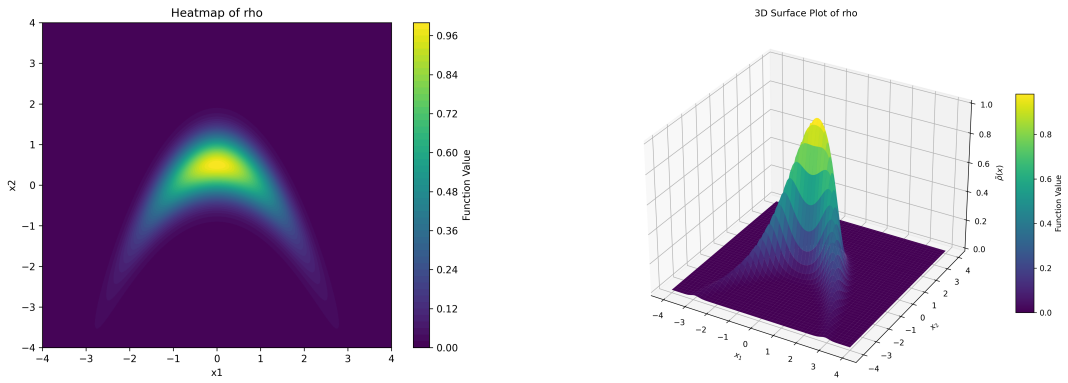
$N = 10^7$, we obtain an estimate of $Z_{\text{large}} = 3.1333$. Comparing this with the estimate $Z_{\text{ref}} = 3.1331$ obtained over $[-4, 4]^2$, we find that the difference is approximately 2×10^{-4}, which is negligible compared to the RMSE of the uniform estimator for all sample sizes considered. This numerical evidence supports the earlier theoretical justification that the truncation error is negligible for practical purposes.

4 Monte Carlo Integration with Importance Sampling

In this section, we implement the Importance Sampling method to estimate the normalizing constant Z.

4.1 Motivation

To better capture the behavior of the integrand $\tilde{\rho}(x)$, we first visualize its heatmap and 3D surface plot over the region $[-4, 4]^2$. Figures 4.1a and 4.1b show the heatmap and 3D surface plot of $\tilde{\rho}(x)$ respectively.



(a) Heatmap of $\tilde{\rho}(x)$

(b) 3D Surface Plot of $\tilde{\rho}(x)$

Figure 4.1: Visualization of the Integrand $\tilde{\rho}(x)$

We can observe from the plots that density $\tilde{\rho}(x)$ is highly concentrated around the curve defined by $2x_2 + x_1^2 - 1 = 0$. Futhermore, the density decays rapidly away from this curve, and is negligible in most regions of the domain. This observation suggests that uniform sampling is inefficient, as many samples fall in low-density regions, contributing little to the estimate of Z. Importance Sampling addresses this inefficiency by drawing samples from a proposal distribution that better matches the shape of the integrand.

4.2 Method and Implementation

4.2.1 Importance Sampling Method

The Importance Sampling estimator for Z is given by

$$\hat{Z}_N^{\text{IS}} = \frac{1}{N} \sum_{i=1}^{N} \frac{\tilde{\rho}(Y_i)}{q(Y_i)}, \quad Y_i \sim q,$$

where Y_i are i.i.d. samples with density function $q(x)$ and N is the total number of samples.

By Strong Law of Large Numbers,

$$\hat{Z}_N^{\text{IS}} \xrightarrow{a.s.} \mathbb{E}\left[\frac{\tilde{\rho}(Y_i)}{q(Y_i)}\right] \quad \text{as } N \to \infty.$$

Here,

$$\mathbb{E}\left[\frac{\tilde{\rho}(Y_i)}{q(Y_i)}\right] = \int_{\mathbb{R}^2} \frac{\tilde{\rho}(y)}{q(y)} q(y)\,\mathrm{d}y = \int_{\mathbb{R}^2} \tilde{\rho}(y)\,\mathrm{d}y = Z.$$

Thus, the Importance Sampling estimator is consistent.

Note that we don't need to restrict the samples Y_i to the region $\Omega = [-4, 4]^2$, since the proposal distribution $q(x)$ has support over the entire $\mathbb{R}^2$.

4.2.2 Algorithm

Similarly, the algorithm for Importance Sampling is as follows:

1. Define the proposal distribution $q(x) = \mathcal{N}(m, \Sigma)$ with mean m and covariance Σ.
2. For a given sample size N, generate N i.i.d. samples $Y_1, Y_2, \ldots, Y_N$ from the distribution $q(x)$.
3. Compute the Importance Sampling estimator

$$\hat{Z}_N^{\text{IS}} = \frac{1}{N} \sum_{i=1}^{N} \frac{\tilde{\rho}(Y_i)}{q(Y_i)}.$$

4. Return the estimate $\hat{Z}_N^{\text{IS}}$.
5. Compute the empirical RMSE by repeating the estimation process multiple times for each N.
6. Repeat the above steps for different values of N to analyze the convergence behavior.

4.3 Choosing Proposal Distribution

Guided by the observed concentration of $\tilde{\rho}(x)$ along the curve $2x_2 + x_1^2 - 1 = 0$, main mass of the density centers around the point $(x_1, x_2) = (0, \frac{1}{2})$. We choose a bivariate normal distribution centered at this point as our proposal distribution:

$$q(x) = \mathcal{N}(m, \Sigma),$$

where m is mean and Σ is covariance matrix.

We next determine appropriate values for m and Σ using the properties of

$$\tilde{\rho}(x) = \exp\left(-\frac{x_1^2}{2} - \frac{1}{2}\left(2x_2 + x_1^2 - 1\right)^2\right).$$

We set the mean as the center of the main mass:

Let

$$\nabla\tilde{\rho} = \left(\frac{\partial\tilde{\rho}}{\partial x_1}, \frac{\partial\tilde{\rho}}{\partial x_2}\right) = 0,$$

where

$$\begin{cases} \dfrac{\partial\tilde{\rho}}{\partial x_1} = \left(-x_1 - 2x_1(2x_2 + x_1^2 - 1)\right)\tilde{\rho}(x), \\ \dfrac{\partial\tilde{\rho}}{\partial x_2} = -2\left(2x_2 + x_1^2 - 1\right)\tilde{\rho}(x). \end{cases}$$

Solving the system yields the critical point at $(x_1, x_2) = (0, \frac{1}{2})$. Therefore, we set

$$m = \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix}.$$

To determine the covariance matrix Σ, we analyze the structure of $\tilde{\rho}$:
Fixing x_2, we can neglect the higher-order term

$$\frac{1}{2}\left(2x_2 + x_1^2 - 1\right)^2.$$

The dominant term in x_1 is then

$$\exp\left(-\frac{x_1^2}{2}\right).$$

Thus, in the x_1 direction, $\tilde{\rho}(x)$ behaves approximately like a standard normal distribution $\mathcal{N}(0, 1)$.

Fixing x_1, we only need to consider the term

$$\exp\left(-\frac{1}{2}\left(2x_2 + x_1^2 - 1\right)^2\right).$$

By substituting $u = \frac{1}{2}\left(x_1^2 - 1\right)$, we see that this term behaves like

$$\exp\left(-2\left(x_2 + u\right)^2\right).$$

This indicates that in the x_2 direction, $\tilde{\rho}(x)$ behaves approximately like a normal distribution $\mathcal{N}\left(-\frac{u}{2}, \frac{1}{2}\right)$.

Since there is little correlation between x_1 and x_2, we select the covariance matrix as

$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix}.$$

This choice captures the main features of $\tilde{\rho}(x)$, allowing for more efficient sampling.

4.4 Numerical Experiments with the Initial Proposal

We conduct numerical experiments to evaluate the performance of the Importance Sampling method.

4.4.1 Setup

We follow the same experimental setup as in Section 3.2.1. We vary the sample size N from 10^2 to 10^7 and compute the estimator $\hat{Z}_N^{\text{IS}}$ for each N. For each sample size, we repeat the estimation process 1000 times to compute the empirical RMSE.

We generate i.i.d. samples $Y_i \sim q(x)$ using NumPy's multivariate normal random number generator.

4.4.2 Experiment Results

Figure 4.2 shows the convergence of the Importance Sampling estimator $\hat{Z}_N^{\text{IS}}$ as the sample size N increases. Unlike the uniform sampling case, the Importance Sampling estimator remain highly unstable. Even at large sample size of $N = 10^7$ it is insufficient to achieve stable estimates.

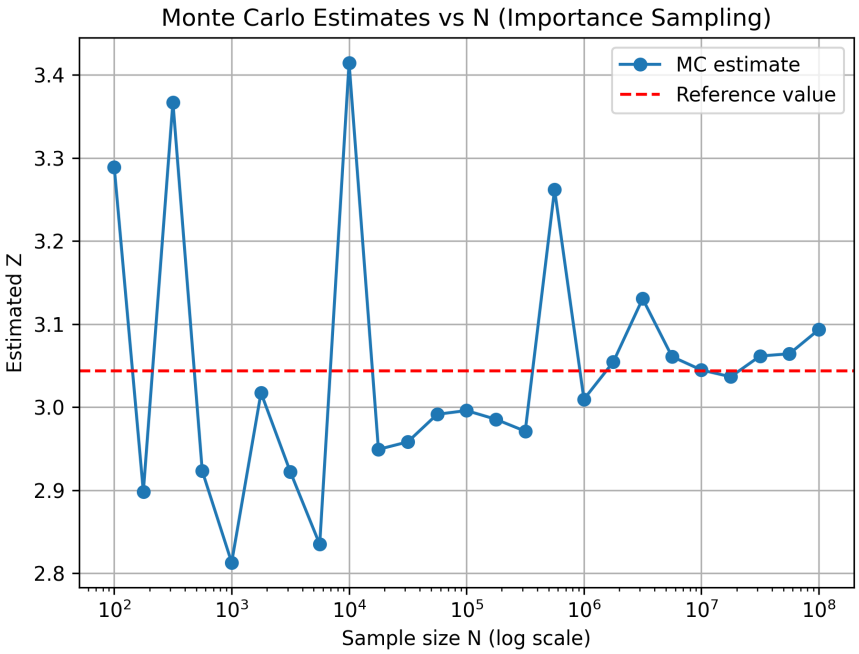


Figure 4.2: Convergence of Importance Sampling Estimator with Initial Proposal

Figure 4.3 presents the RMSE of the Importance Sampling estimator as sample size N on a log-log scale. A log-log regression yields a slope of approximately -0.13, substantially slower than the theoretical Monte Carlo rate of -0.5. Significant oscillations in RMSE persist even at large sample sizes, indicating severe instability.

These results demonstrate that the initial Gaussian proposal fails to provide an efficient importance sampling estimator.

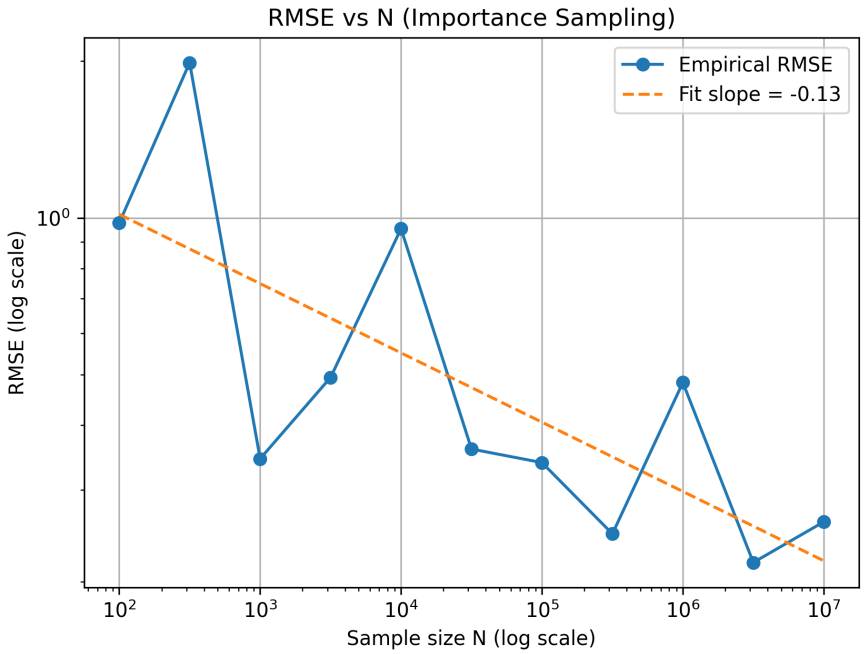


Figure 4.3: RMSE of Importance Sampling Estimator with Initial Proposal vs Sample Size

4.4.3 Discussion

In this section we interpret the numerical results and explain why the initial importance sampling procedure performs poorly.

The poor numerical performance of the initial importance sampling estimator can be explained by the geometric structure of the target density

$$\tilde{\rho}(x) = \exp\left(-\frac{x_1^2}{2} - \frac{1}{2}\left(2x_2 + x_1^2 - 1\right)^2\right).$$

Although the density has a unique mode at $(0, \frac{1}{2})$, it is highly concentrated along the curved manifold defined by $2x_2 + x_1^2 - 1 = 0$. The initial Gaussian proposal $q(x) = \mathcal{N}(m, \Sigma)$ with mean $m = (0, 0.5)$ and covariance $\mathrm{diag}(1, 0.25)$ fails to adequately capture this curved structure. As a result, many samples drawn from $q(x)$ fall in regions where $\tilde{\rho}(x)$ is negligible. This leads to extremely high variance in the importance weights

$$w(x) = \frac{\tilde{\rho}(x)}{q(x)},$$

resulting in an unstable estimator.

The heatmap of the importance weights $w(x)$, shown in Figure 4.4, confirms this analysis. The weights exhibit extreme variability, with large regions of near-zero values and isolated spikes of very high values. This indicates that the proposal distribution does not align well with the target density, leading to inefficient sampling and poor estimator performance.

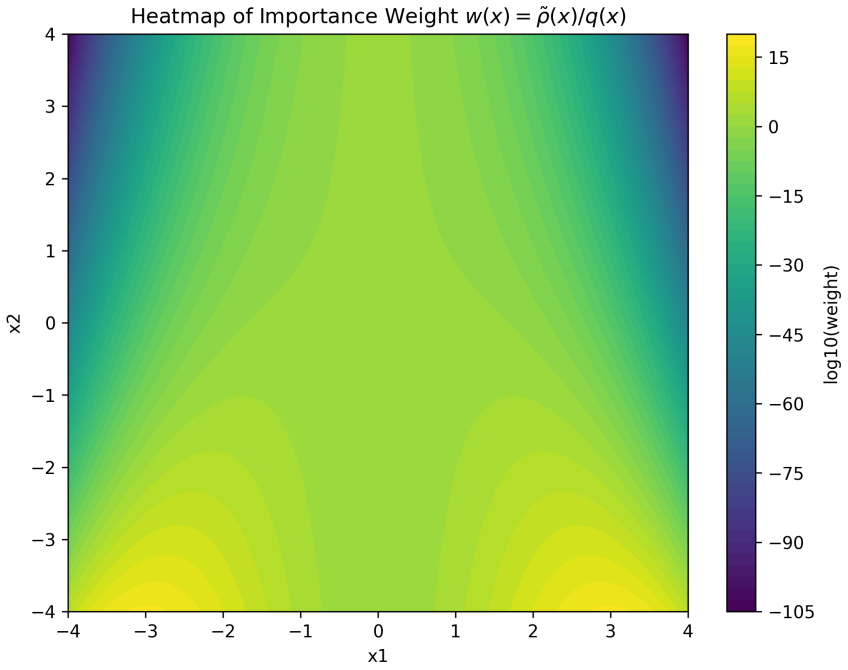


Figure 4.4: Heatmap of Initial Importance Weights $w(x) = \tilde{\rho}(x)/q(x)$ in log scale

4.5 Refined Proposal Distribution and Numerical Experiments

The numerical results in Section 4.4 suggest that the main issue of the initial proposal is not its local curvature near the mode, but rather its global mismatch with the ridge-like structure of the target density. In this section we explore a simple improvement within the Gaussian family by tuning only the mean of the proposal while keeping the covariance fixed.

4.5.1 Tuning Strategy

To better align the proposal distribution with the ridge of high density, we consider the following two-parameter family of Gaussian proposals:

$$q(x) = \mathcal{N}\left(\tilde{m}, c^2\Sigma\right),$$

where

$$\tilde{m} = \begin{pmatrix} 0 \\ b \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix},$$

and $b \in [0, 1], c > 0$ are tuning parameters.

The choice of this family is motivated by the following considerations:

- Shifting the mean in the x_2 direction allows the proposal to better capture the vertical position of the ridge defined by $2x_2 + x_1^2 - 1 = 0$.

- Scaling the covariance matrix adjusts the spread of the proposal, allowing for more or less concentration around the ridge.
- Empirically, the heatmap of $\tilde{\rho}(x)$ suggests that there is no strong correlation between x_1 and x_2, justifying the diagonal covariance structure.

4.5.2 Tuning Procedure

To identify optimal values for the tuning parameters b and c, we perform a grid search over a reasonable range of values.

As the heatmap of $\tilde{\rho}(x)$ suggests that the ridge lies roughly between $x_2 = 0$ and $x_2 = 1$, we vary b from 0 to 1 in increments of 0.1. For the scaling parameter c, we vary it from 0.5 to 4.5 in increments of 0.1. For each pair (b, c), we run the Importance Sampling estimator with a moderate sample size of $N = 10^4$, repeating the estimation 200 times to compute the empirical RMSE.

By analyzing the RMSE results across the grid, we identify the pair $(b^*, c^*) = (0.2, 2.3)$ as yielding the lowest RMSE. In order to get a more precise estimate of the optimal parameters, we perform a finer grid search around this point with increments of 0.01, ultimately selecting $(b^*, c^*) = (0.19, 2.28)$.

The result makes intuitive sense:

- The optimal mean shift $b^* = 0.19$ positions the proposal closer to the ridge of high density, improving sample efficiency.
- The scaling factor $c^* = 2.28$ increases the spread of the proposal, allowing it to better cover the elongated shape of the target density along the ridge.

4.5.3 Experiment Results with the Refined Proposal

We repeat the numerical experiments from Section 4.4 using the refined proposal distribution

$$q(x) = \mathcal{N}\left(\tilde{m}^*, (c^*)^2\Sigma\right),$$

where

$$\tilde{m}^* = \begin{pmatrix} 0 \\ 0.19 \end{pmatrix}, \quad c^* = 2.28, \quad \Sigma = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix}.$$

Figure 4.5 shows the convergence of the Importance Sampling estimator $\hat{Z}_N^{\mathrm{IS}}$ with the refined proposal as the sample size N increases.

Figure 4.6 presents the RMSE of the Importance Sampling estimator as sample size N on a log-log scale. A log-log regression yields a slope of approximately -0.42, closer to the theoretical Monte Carlo rate of -0.5.

Figure 4.7 shows a scatter plot of 5000 samples drawn from the refined proposal distribution.

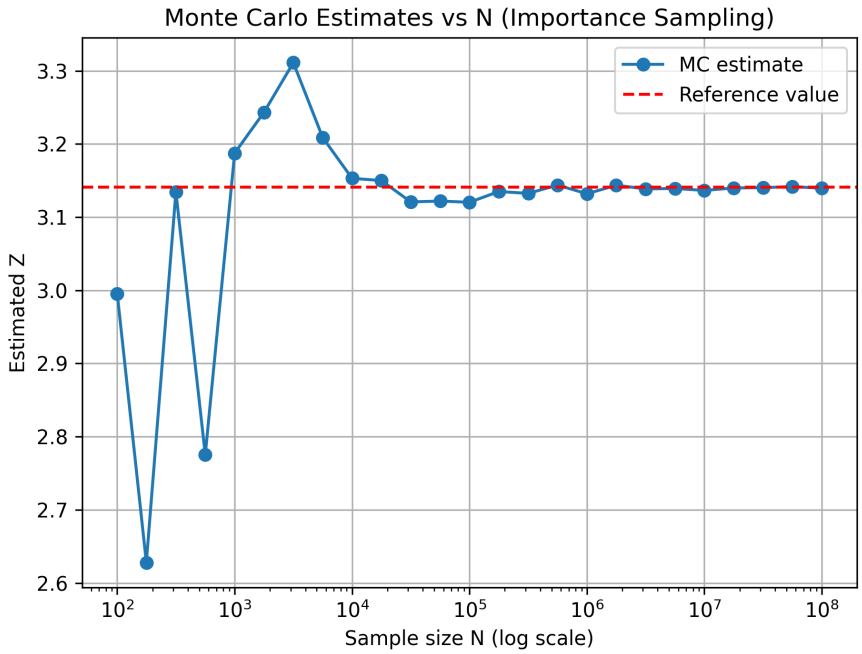


Figure 4.5: Convergence of Importance Sampling Estimator with Refined Proposal

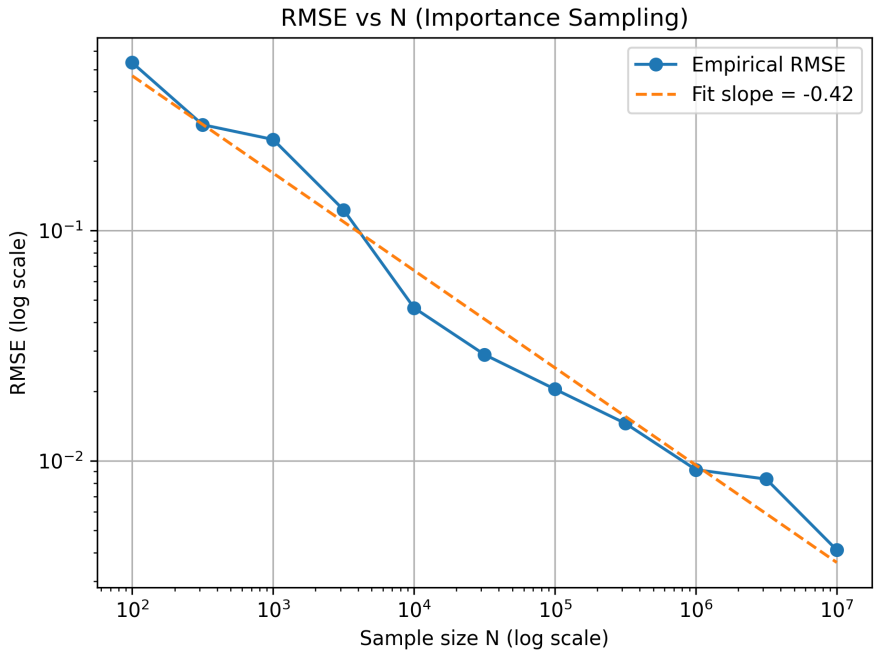


Figure 4.6: RMSE of Importance Sampling Estimator with Refined Proposal vs Sample Size

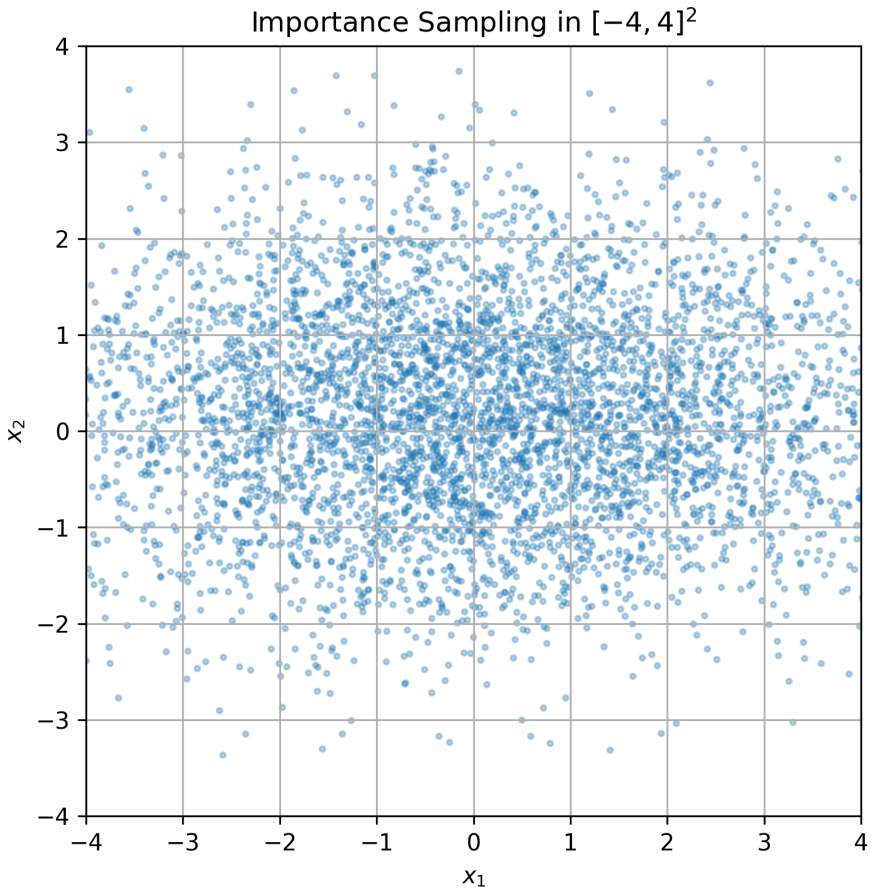
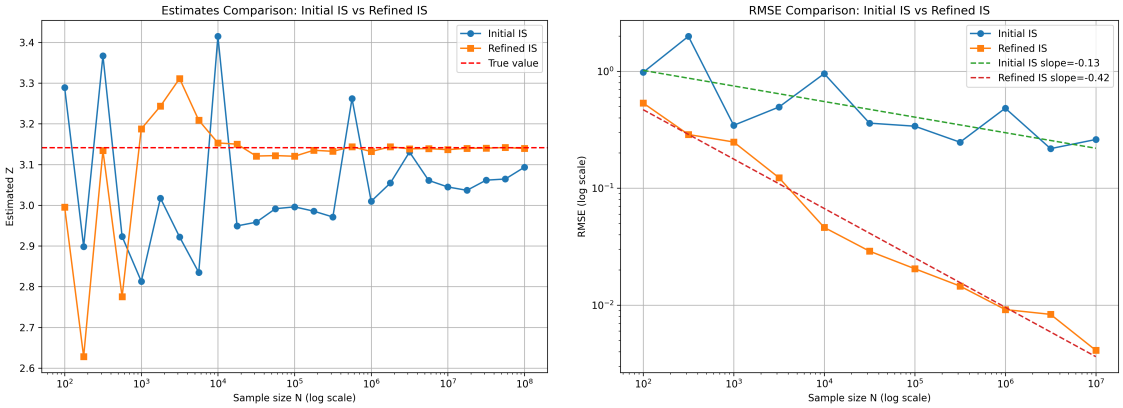


Figure 4.7: Sample Distribution of Importance Sampling with Refined Proposal ($N = 5000$)

4.5.4 Discussion of Refined Proposal

The refined proposal distribution leads to a clear improvement in the performance of the Importance Sampling estimator. Unlike the initial isotropic Gaussian proposal, the refined proposal shifts the mean in the x_2 direction and increases the variance along the ridge of the target density, resulting in a noticeably better overlap between the proposal density q and the target density ρ.

This structural alignment yields two important effects. First, the variance of the importance weights is significantly reduced, which in turn makes the estimator substantially more stable. This improvement is particularly visible in the convergence plot of Figure 4.8a, where the refined estimator stays close to the true value even for moderate sample sizes, whereas the initial proposal exhibits large fluctuations. This is also confirmed by the RMSE comparison in Figure 4.8b: for all sufficiently large N, the refined proposal yields a uniformly lower RMSE.



(a) Comparison of Convergence

(b) Comparison of RMSE

Figure 4.8: Comparison of Importance Sampling with Initial and Refined Proposals

In contrast, the heatmap for the refined proposal shown in Figure 4.9 demonstrates a much more uniform distribution of importance weights, with fewer extreme values and a noticeably smaller dynamic range. This visual evidence supports the quantitative findings: the refined proposal better captures the geometry of the target density and avoids the degeneracy inherent in the initial proposal.

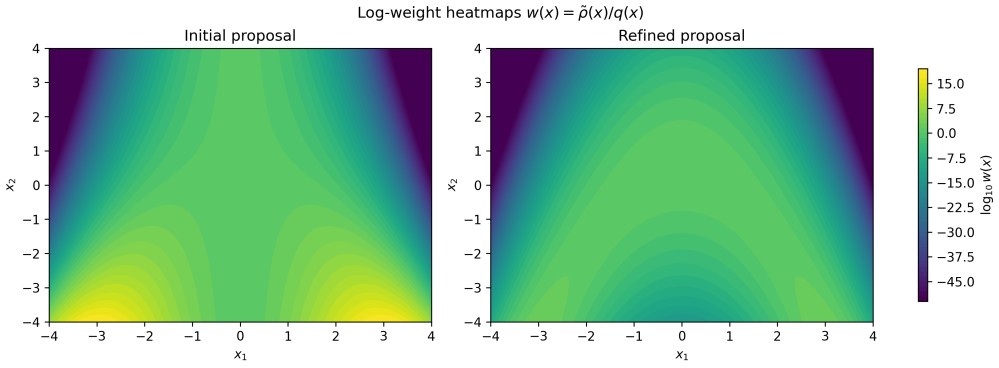


Figure 4.9: Heatmap of Refined Importance Weights $w(x) = \tilde{\rho}(x)/q(x)$ in log scale

4.6 Comparison between Uniform Sampling and Importance Sampling

Finally, we compare the performance of the uniform sampling Monte Carlo estimator and the Importance Sampling estimator with the refined proposal.

Since the integration domain is compact and the target density $\tilde{\rho}(x)$ is smooth and does not exhibit extreme concentration, the uniform estimator is already reasonably efficient for this problem. This explains why its RMSE curve in Figure 4.10 displays a clean $N^{-1/2}$ decay with a relatively small variance constant.

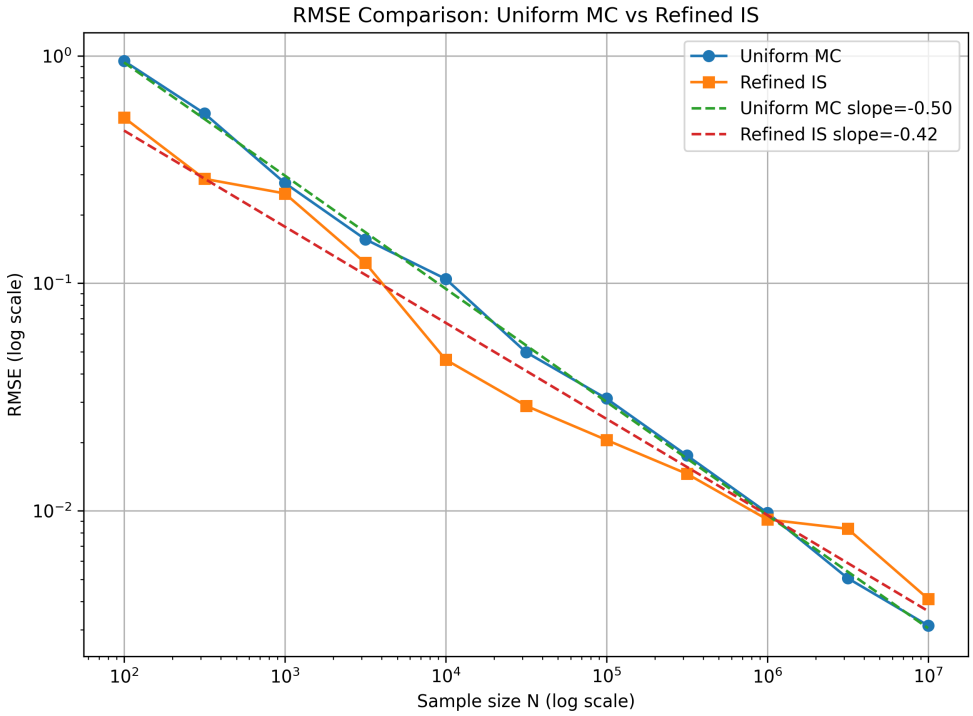


Figure 4.10: Comparison of RMSE between Uniform Sampling and Importance Sampling Estimators

Nevertheless, the Importance Sampling estimator still outperforms the uniform estimator across all sample sizes. The convergence plot in Figure 4.11 shows that the Importance Sampling estimator remains more stable compared to the uniform sampling estimator. Note that the uniform sampling estimator is slightly biased due to the truncation to $[-4, 4]^2$.

Although the improvement over uniform sampling is not dramatic, the refined proposal nonetheless offers a measurable and robust variance reduction. This highlights the value of adapting the proposal distribution to the structure of the target, even when the baseline Monte Carlo estimator is already reasonably effective.

5 Conclusion

In this report, we have explored Monte Carlo Integration methods for estimating the normalizing constant of a given unnormalized density function $\tilde{\rho}(x)$ over $\mathbb{R}^2$.

We began by justifying the restriction of the integration domain to a compact region $[-4, 4]^2$, demonstrat-

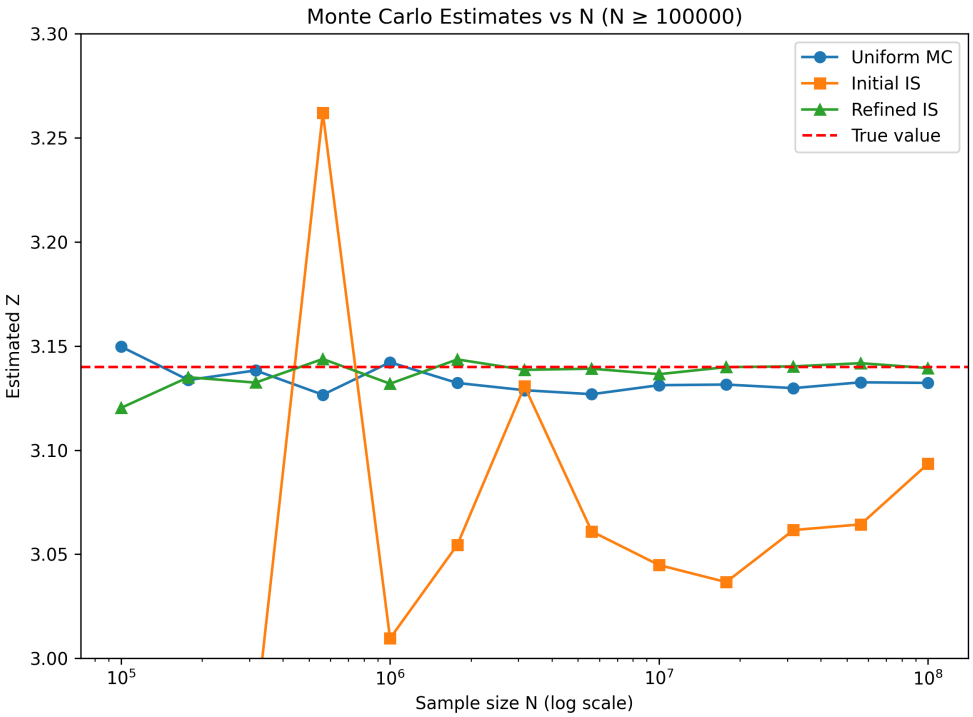


Figure 4.11: Comparison of Uniform Sampling and Importance Sampling Estimators

ing that the tail contribution is negligible compared to the statistical error of the estimators. We then implemented the standard Monte Carlo Integration method using uniform sampling, analyzing its convergence properties and empirical performance through numerical experiments. Subsequently, we introduced the Importance Sampling method to improve estimation efficiency. We designed an initial Gaussian proposal distribution based on the local curvature of $\tilde{\rho}(x)$, but found that it failed to capture the global structure of the density, leading to poor performance. To address this, we refined the proposal by tuning its mean and covariance to better align with the ridge-like structure of $\tilde{\rho}(x)$. This refinement resulted in significant improvements in estimator stability and accuracy, as confirmed by numerical experiments.

Due to time limitations, we focused on basic Monte Carlo methods and Gaussian proposals. Future work could explore more advanced techniques such as Markov Chain Monte Carlo (MCMC) methods.

Overall, this report highlights the importance of tailoring sampling strategies to the geometry of the target density in order to achieve efficient and accurate Monte Carlo integration.